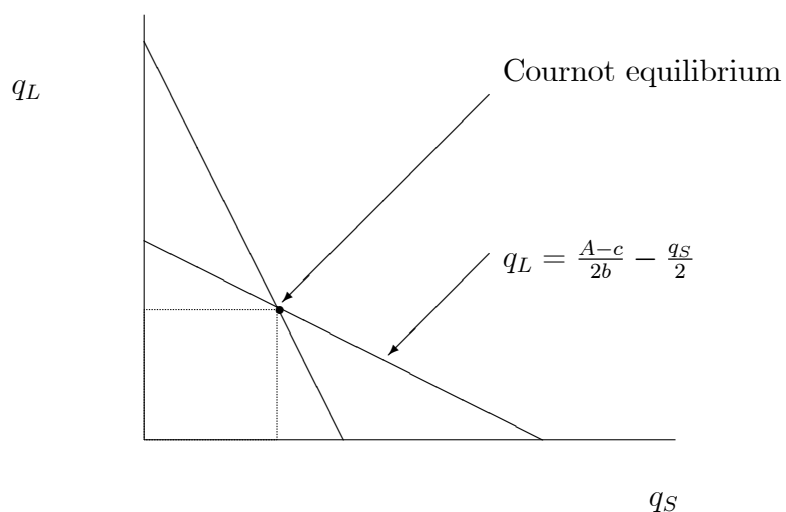


Imperfect Competition II: Oligopoly

1 Oligopoly

1.1 Cournot oligopoly model

- Two-firm Cournot model: $P = A - b(q_S + q_L)$, and $MC_S = MC_L = c$.



- Reaction function

$$q_S = \frac{A - c}{2b} - \frac{q_L}{2},$$

$$q_L = \frac{A - c}{2b} - \frac{q_S}{2}.$$

- N -firm Cournot model

$$P = A - b \sum_{i=1}^N q_i.$$

Cost function for each firm

$$C(q_i) = \alpha q_i^2 + \beta q_i + \theta.$$

Here $\alpha > 0$, $\beta > 0$ and $A > \beta$.

4. Each firm i maximizes profit by choosing q_i

$$\Pi_i(N) = \left(A - b \sum_{j=1}^N q_j \right) q_i - (\alpha q_i^2 + \beta q_i + \theta_i).$$

which gives the reaction function

$$q_i = \frac{A - \beta - b \sum_{j \neq i} q_j}{(2b + 2\alpha)}.$$

5. In equilibrium each firm i produces an output

$$q^* = \frac{A - \beta}{b(N + 1) + 2\alpha},$$

and the market price is given by

$$P^* = \frac{bA + 2\alpha A + bN\beta}{b(N + 1) + 2\alpha}.$$

The profit for each firm

$$\Pi^i(N) = \frac{(A - \beta)^2(b + \alpha)}{[b(N + 1) + 2\alpha]^2} - \theta. \quad (1)$$

1.2 Bertrand model

1. Bertrand model with homogeneous product

$$q_i = \begin{cases} 0 & P_i > \max_{j \neq i} \{P_j\} \\ q & P_i < \min_{j \neq i} \{P_j\} \\ \frac{q}{h} & P_i = \min \{P_j\}, \quad h = \#\{P_k = \min \{P_j\}\} \end{cases}$$

Suppose all firms have same marginal cost c , the unique Bertrand equilibrium, $P_i = c$ for all i .

2. Heterogeneous products: n identical firms, $MC = c$

$$q_i = A - P_i - \gamma \left(P_i - \frac{1}{n} \sum_{j=1}^n P_j \right)$$

The Bertrand equilibrium price:

$$P_i^* = \frac{A + c(1 + \frac{\gamma(n-1)}{n})}{2 + \frac{\gamma(n-1)}{n}}.$$

Here γ can be understood as a substitutability coefficient: $\gamma = 0$, products are non-substitutes; each firm acts as monopolist in the market. $\gamma = \infty$, products are perfect substitutes (homogeneous).

Strategies that can increase profits: Advertisement stressing differences between own product and those of competing firms, brand name building, increasing customer loyalty

1.3 The Stackelberg model of oligopoly

1. The Stackelberg model of oligopoly pertains to a situation in which one firm acts as a quantity leader, choosing its quantity first, with all other firms acting as followers, making their quantity decision after the leader has moved.
2. Example: Samsung as leader, LG as follower

$$P = 100 - q, \quad MC = 10.$$

LG takes q_S as given

$$q_L = 45 - \frac{q_S}{2}.$$

For Samsung, knowing that LG maximizes profit this way,

$$P = 100 - q_S - \left(45 - \frac{q_S}{2}\right) = 55 - \frac{q_S}{2}.$$

Maximizing profit with respect to q_S gives

$$q_S = 45.$$

Plugging q_S into the reaction function of q_L gives $q_L = 22.5$.

1.4 Dominant firm market

1. Dominant firm markets are markets in which one firm possesses a large share of the markets but competes against numerous small firms, each offering identical products.
2. Market structure: dominant firm sets the price (price setter); fringe firms act as perfect competitors (price takers)

3. Example 1: One dominant firm with constant market cost $MC = 25$, and one fringe firm with supply curve

$$q_f = P - 25.$$

Market demand is given by

$$P = 125 - Q$$

For dominant firm

$$P = (125 - q_f) - q_d \implies P = 75 - 0.5q_d.$$

Profit-maximizing condition

$$MR = 75 - q_d = MC = 25 \implies q_d = 50, \quad q_f = 25, \quad P = 50.$$

4. Example 2: One dominant firm with cost function

$$C_d = 50 + 10q_d + \frac{1}{6}q_d^2.$$

There are 8 identical fringe firms, each of which has cost function

$$C_f = 10 + 2q_f^2.$$

Market demand $P = 390 - Q$. The equilibrium

$$q_d = 120, \quad q_f = \frac{45}{2}, \quad P = 90.$$

1.5 Monopolistic competition

1. Firm's problem

$$\max_{P^j} \pi^j(P) \equiv q^j(P)P^j - c^j(q^j(P)), \quad P = (P^1, \dots, P^j, \dots).$$

2. 3 features:

- (a) Many buyers and sellers, but market is fragmented.
- (b) Free entry and exit.
- (c) Firms produce horizontally differentiated products.

3. Key difference between monopolistic competition and differentiated product oligopoly: free entry.
4. In long-run, monopolistic competing firms make **zero** economic profit.
5. Price may not fall as more firms enter the market.

2 Repeated Interactions

2.1 The Setup

1. The two firms produce perfect substitutes with the same marginal cost, c . The low-price firm gets the whole market, and the firms share the market when they charge the same price. The Bertrand game play $T + 1$ times.
2. At each date t , the firms choose their prices (p_{1t}, p_{2t}) simultaneously. There is no physical link between the periods; the previous choice of price by one's rival is already obsolete when one chooses a price. Nevertheless, we will allow price choices at date t to depend on the history of previous prices. Thus, the price strategy p_{it} depends on the history

$$H_t \equiv (p_{1,0}, p_{2,0}; \dots; p_{1,t-1}, p_{2,t-1}).$$

3. We require strategies to form a "public perfect equilibrium". That is for any public history H_t at date t , firm i 's strategy from date t on maximizes the present discounted value of profit given firm j 's strategy from that date on.
4. First we assume that the horizon is finite: $T < \infty$. Hence, the equilibrium is, for any history, the repeated Bertrand one. Therefore, the dynamic element contributes nothing to the model.

2.2 The Tacit Collusion

1. The picture changes dramatically when the horizon is infinite: $T = \infty$. On one hand, it is easy to verify that the Bertrand equilibrium repeated infinitely is an equilibrium of this game. To see this, consider the following strategy: each firm chooses a price equal to the marginal cost in each period t , regardless of the history of the game up to t . Given that the rival firm charges a price equal to c in this manner, each firm can do no better than to charge c itself.
2. On the other hand, the interesting feature of this game is that the repeated Bertrand equilibrium is no longer the only equilibrium. Let p^m be the monopoly price (it maximizes $(p - c)D(p)$) and consider the following (symmetric) strategies: each firm charges p^m in period 0. It furthermore charges p^m in period t if in every period preceding t both firms have charged p^m ; otherwise it sets its price at marginal cost c forever.

3. Formally: $p_{it}(H_t) = p^m$ if $H_t = (p^m, p^m, \dots, p^m)$; and $p_{it}(H_t) = c$ otherwise. These strategies are called trigger strategies, because a single deviation triggers a halt in the cooperation. They constitute an equilibrium if the discount factor is sufficiently high: in charge p^m , a firm earns half the monopoly profit in each period. By deviating from this price, a firm can earn maximum profit, π^m , during the period of deviation, but then it receives zero forever after.
4. Therefore if

$$\frac{\pi^m}{2}(1 + \delta + \delta^2 + \dots) \geq \pi^m$$

which follows if $\delta \geq \frac{1}{2}$, then these trigger strategies are equilibrium ones. This result is a formalization of tacit collusion. Note that collusion is enforced through a purely noncooperative mechanism.

2.2.1 The Folk Theorem

1. There are many other equilibria in this game. The previous reasoning actually implies that any price between the competitive price and the monopoly price can be sustained as an equilibrium as long as the discount factor is greater than $\frac{1}{2}$ (which implies any symmetric per-period profit between 0 and π^m can be an equilibrium profit).
2. Let $p \in [c, p^m]$. and let each firm charge price p as long as neither has yet deviated from that price. If either firm has deviated in the past, both charge the competitive price forever. Again, these strategies are equilibrium strategies. By conforming to p , each firm gets

$$\frac{\pi(p)}{2}(1 + \delta + \delta^2 + \dots).$$

3. If a firm deviates, it gets at most $\pi(p)$ during the deviation period, and it loses the profit at price p forever after. Thus if

$$\frac{\pi(p)}{2}(1 + \delta + \delta^2 + \dots) \geq \pi(p),$$

that is, if $\delta \geq \frac{1}{2}$, deviating from price p is not privately optimal.

4. The result is one facet of a general result, known as the Folk theorem. For the repeated game under consideration, the Folk theorem asserts that any pair of profits (π^1, π^2) such that

$$\pi^1 > 0, \pi^2 > 0, \text{ and } \pi^1 + \pi^2 \leq \pi^m$$

is a per-period equilibrium payoff for δ sufficiently close to 1.

5. That is, there exists perfect equilibrium strategies

$$\{p_{1t}(H_t), p_{2t}(H_t)\}$$

that form a perfect equilibrium such that, for all i , firm i 's associated per-period payoff

$$(1 - \delta) \sum_{t=0}^{\infty} \delta^t \pi^i(p_{it}, p_{jt}) = \pi^i.$$

6. We can draw the attainable per-period payoffs for δ close to 1.

2.3 Market Concentration

1. Bain's (1956) original concern with market concentration was based on an intuitive relationship between high concentration and collusion. There is a sense in which tacit collusion is easier to sustain with a smaller number of firms.
2. Consider a homogenous-good industry with n firms facing the same constant marginal cost, and look at the fully collusive outcome, in which all firms charge the monopoly price and share the market equally.
3. The per period and per-firm profit is $\frac{\pi^m}{n}$, a decreasing function of n . A large number of firms reduces the profit per firm and thus the cost of being punished for undercutting. In contrast, the short-run gain from undercutting the monopoly price slightly is

$$\pi^m \left(1 - \frac{1}{n}\right) - \varepsilon$$

and thus increase with n .

4. The discount factor must exceed $1 - \frac{1}{n}$ for collusion to be sustainable, in this sense, market concentration facilitates tacit collusion.

2.4 Long Information Lags and Infrequent Interaction

1. The threat of a punishment operates only if the punishment comes fairly soon after a price cut. Punishment might be delayed for two related reasons. First, a firm's price cut may be learned of by its rival only with a lag. Second, infrequent interaction (due to lumpiness in orders, for instance) delays the punishments and makes current price cutting more attractive.

2. A more infrequent interaction corresponds to a decrease in δ . To formalize the first reason, we assume that prices are observed two periods (instead of one) after they are chosen. Assume further that a firm's profit and demand in a given period are observed by this firm at least two periods later, so that a firm cannot infer from the observation of its past profits and demands anything it does not know about its rival's pricing behavior.
3. In such a situation, a firm can deviate and cut its price for two periods before being detected. The monopoly price is sustainable in equilibrium if and only if

$$\frac{\pi^m}{2}(1 + \delta + \delta^2 + \dots) \geq \pi^m(1 + \delta)$$

or $\delta \geq \frac{1}{\sqrt{2}}$.

4. Thus, the condition is more stringent than the no lag detection case, where $\delta \geq \frac{1}{2}$. In this sense, information lags are also a cause of breakdown of collusion.

2.5 Fluctuating Demand

1. Let us now consider Rotemberg and Saloner's (1986) theory of price war during booms. Suppose that demand is stochastic. At each period t , it can be low ($q = D_1(p)$) with prob. $\frac{1}{2}$ or high ($q = D_2(p)$) with prob. $\frac{1}{2}$, where $D_2(p) > D_1(p)$ for all p .
2. Assume that the demand shock is identically and independently distributed over time. At each period the two firms learn the current state of demand before choosing their prices simultaneously.
3. We look for a pair of prices $\{p_1, p_2\}$ such that (a) both firms charge prices p_s when the state of demand is s , (b) the price configuration $\{p_1, p_2\}$ is sustainable in equilibrium, and (c) the expected present discounted profit of each firm along the equilibrium path is not Pareto dominated by other equilibrium payoffs.
4. The expected present discounted profit is

$$V = \frac{1}{2} \frac{D_1(p_1)}{2} (p_1 - c) + \frac{1}{2} \frac{D_2(p_2)}{2} (p_2 - c) + \delta V.$$

We know from the maximum-punishment principle that in order to enforce the pair of price $\{p_1, p_2\}$, one can assume that after a deviation the two firms charge their competitive price c forever.

2.5.1 Fully collusive outcome

1. By "fully collusive outcome" we mean that the two firms charge the monopoly price p_s^m in each state. Let $\pi_s^m = (p_s^m - c)D_s(p_s^m)$ denote the monopoly profit in state s . If the monopoly can always be sustained, then

$$V = \frac{\pi_1^m + \pi_2^m}{4} + \delta V. \text{ or } V = \frac{\pi_1^m + \pi_2^m}{4(1 - \delta)}$$

2. From the maximum-punishment principle, the future loss from deviating is δV . Undercutting p_s^m slightly in state s yields an extra gain of almost

$$\pi_s^m - \frac{\pi_s^m}{2} = \frac{\pi_s^m}{2}$$

to the deviating firm. Thus for p_s^m to be sustainable for all s , one must have

$$\frac{\pi_s^m}{2} \leq \delta V \quad (2)$$

3. Because $\pi_1^m < \pi_2^m$, this condition is satisfied iff

$$\frac{\pi_2^m}{2} \leq \delta V \quad (3)$$

or substituting for V ,

$$\delta \geq \delta_0 \equiv \frac{2\pi_2^m}{3\pi_2^m + \pi_1^m}. \quad (4)$$

Because $\pi_1^m < \pi_2^m$, δ_0 lies strictly between $\frac{1}{2}$ and $\frac{2}{3}$.

4. This result already offers some insight. When demand is high, the temptation to undercut is stronger. The interesting case is that $\delta \in [\frac{1}{2}, \delta_0)$. We must choose $\{p_1, p_2\}$ so as to maximize the firms' expected payoff subject to the incentive constraints:

$$\max \left\{ \frac{1}{2} \frac{\pi_1(p_1)}{2} + \frac{1}{2} \frac{\pi_2(p_2)}{2} \right\} / (1 - \delta) \quad (5)$$

subject to

$$\frac{\pi_1(p_1)}{2} \leq \delta \left(\frac{1}{2} \frac{\pi_1(p_1)}{2} + \frac{1}{2} \frac{\pi_2(p_2)}{2} \right) / (1 - \delta) \quad (6)$$

$$\frac{\pi_2(p_2)}{2} \leq \delta \left(\frac{1}{2} \frac{\pi_1(p_1)}{2} + \frac{1}{2} \frac{\pi_2(p_2)}{2} \right) / (1 - \delta). \quad (7)$$

5. Intuitively, the binding constraint will be (7) since the temptation to undercut is higher when demand is high. Indeed, it can easily be shown that this is the case. Note that the program is equivalent to

$$\max [\pi_1(p_1) + \pi_2(p_2)]$$

subject to

$$\pi_1(p_1) \leq K\pi_2(p_2) \tag{8}$$

$$\pi_2(p_2) \leq K\pi_1(p_1) \tag{9}$$

where

$$K \equiv \frac{\delta}{2 - 3\delta} \geq 1.$$

6. Ignore (8) and solve the above program. Clearly, choosing $p_1^* = p_1^m$ increase the objective function and relaxes constraint (9) as much as possible. And choose $p_2^* < p_2^m$ such that $\pi_2(p_2^*) = K\pi_1(p_1^m) = K\pi_1^m$.
7. We thus conclude the following: for $\delta \in [\frac{1}{2}, \delta_0)$, some collusion is sustainable. In the low state of demand, firms charge the monopoly price in that state. In the high state of demand, firms charge below the monopoly price in that state. Rotemberg and Saloner interpret this as demonstrating the existence of a price war during the booms, where firms are forced to lower the amount of their collusion in good times.

2.6 Secret Price Cuts

1. In the previous models, a firm's past price choice is perfectly observed by its rival. However, one can think of instances in which prices are not observable. A firm may offer a price discount to a customer. If its rival's price is unobservable, a firm must rely on the observation of its own market share or demand to detect any price undercutting by the rival.
2. However, if the demand function is random and shocks are unobservable, this inference process is garbled. A low market share may be due to the aggressive behavior of one's rival or to a slack in demand. Hence, when demand is very random, price cuts are hard to detect. As Stigler (1964) noted, this tends to hinder collusion.
3. Green and Porter (1984) have developed a supergame model to formalizes the issue of secret price cutting. In their model firms charge the monopoly price as long as their profit has been high in

the past. If a firm observes a low profit, which could be due to a price cut by its rival or to a low demand, or if itself has undercut the monopoly price in the past period, it charges a low price for some period of time T , so does its rival (this is a punishment phase). The firms revert to the collusive phase, that is, charge the monopoly price, after the punishment phase is completed, until the next deviation or slump in demand.

4. The model then predicts periodic price wars, contrary to the perfect observation models discussed above. Price wars are involuntary, in that they are triggered not by a price cut but by an unobservable slump in demand. Note also that price wars are triggered by a recession, contrary to the Rotemberg-Saloner model.